Chain-of-Thought Prompting: Eliciting Reasoning in LLMs

FAMAF - UNC

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🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

🧉 Chain-of-Thought Prompting Elicits Reasoning in Large Language Models ---

(Wei, Wang, et al. 2022)

🧉 Introduction & Related Work ---

🧉 Challenges in LLMs ---

- Scaling up model size alone has not proved sufficient for achieving high performance on challenging tasks such as arithmetic, commonsense, and symbolic reasoning.
- Large language models still have limitations in their ability to reason and understand the context of a situation.

Previous Work ---

- Models can generate intermediate natural language steps by training from scratch (Ling et al. 2017) or finetuning a pretrained model (Cobbe et al. 2021).
  - Costly to create large sets of high-quality rationales.
- Large language models offer the prospect of in-context few-shot learning via prompting (Brown et al. 2020).
  - Works poorly on tasks that require reasoning abilities.
  - Often does not improve substantially with increasing model scale.

🧉 Contribution ---

- Explores few-shot prompting for reasoning tasks using prompts structured as triples: $\langle \text{input}, \textit{chainofthought}, \text{output} \rangle$.
  - **Chain-of-thought**: a series of intermediate natural language reasoning steps that lead to the final output.
- Presents empirical evaluations on arithmetic, commonsense, and symbolic reasoning benchmarks, showing that chain-of-thought prompting outperforms standard prompting.

(Wei, Wang, et al. 2022)

🧉 Methodology ---

🧉 How Do We Solve Complicated Problems? ---

Decompose the problem into intermediate steps and solve each before giving the final answer.

Example

"After Jane gives 2 flowers to her mom she has 10 . . . then after she gives 3 to her dad she will have 7 . . . so the answer is 7."

The paper shows sufficiently large LMs can generate chains of thought if demonstrations of chain-of-thought reasoning are provided as few-shot prompting.

🧉 Standard vs Chain-of-Thought Prompting ---

🧉 Benefits of Chain-of-Thought ---

1. Allows models to decompose multi-step problems into intermediate steps, allocating more computation where needed.
2. Provides an interpretable window into the model's reasoning process.
3. Potentially applicable to any task requiring a multi-step approach.
4. Can be induced simply by including chain-of-thought examples in the prompt.

(Wei, Wang, et al. 2022)

🧉 Sensitivity to Prompt Engineering ---

Chain-of-thought is robust to:

1. Different annotators writing prompts.
2. Annotators without machine learning background.
3. Different types, ordering, and number of prompt examples.
4. Different language models (LaMDA, GPT-3, PaLM).

However, prompt engineering still improved performance significantly in many cases.

Robustness: Number of Exemplars ---

🧉 Experiments & Results ---

🧉 Experimental Setup -- Benchmarks ---

Benchmarks (five math word problem datasets):

- GSM8K (Cobbe et al. 2021)
- SVAMP (Patel, Baral, and Baral 2021)
- ASDiv (Miao, Liang, and Su 2020)
- AQuA
- MAWPS (Koncel-Kedziorski et al. 2016)

🧉 Experimental Setup -- Models ---

Baseline: standard few-shot prompting (Brown et al. 2020).

Chain-of-thought prompting tested on five LLMs:

- GPT-3 (Brown et al. 2020)
- LaMDA (Thoppilan et al. 2022)
- PaLM
- UL2 20B (Tay et al. 2022)
- Codex (Chen et al. 2021)

Results -- Arithmetic Reasoning ---

Three key takeaways:

- Chain-of-thought prompting is an **emergent ability of model scale**.
- Larger performance gains for **more-complicated problems**.
- GPT-3 175B and PaLM 540B **compare favorably** to prior state of the art.

🧉 Ablation Study ---

Three variations compared against full chain-of-thought:

- Equation only
- Variable compute only
- Chain-of-thought after the answer

Robustness of Chain-of-Thought ---

Findings: chain-of-thought prompting for arithmetic reasoning is robust to:

- Annotators
- Independently-written chains of thought
- Different exemplars and exemplar orders
- Various language models
- Varying numbers of exemplars

🧉 Commonsense Reasoning ---

🧉 Commonsense & Symbolic: Example Prompts ---

Results -- Commonsense Reasoning ---

- Chain-of-thought prompting works better when the model is **sufficiently large**.

🧉 Symbolic Reasoning ---

Results -- Symbolic Reasoning (OOD) ---

- **In-domain and out-of-domain** evaluations show CoT enables length generalization.

🧉 Limitations & Discussion ---

Limitations

- CoT emulates human reasoning but does not prove the network actually "reasons."
- Emergence only at large model scales makes deployment costly.

Discussion Questions

- What reasoning tasks may not be well-suited to chain-of-thought?
- What other prompting methods might expand LM capabilities?
- Does chain-of-thought truly improve interpretability?

🧉 Can Language Models Learn from Explanations in Context? ---

(Lampinen et al. 2022)

Introduction ---

- Language models can perform new tasks by adapting to a few in-context examples (**few-shot learning**) (Brown et al. 2020).
- **Key question:** Could including few-shot *explanations* of the answers in these examples improve LM performance?

🧉 Setup: Training vs Evaluation ---

🧉 Related Work ---

- **In-context learning**: Min et al. (2022) show ground truth labels matter less than label space/distribution; Webson & Pavlick (2021) question whether models truly understand prompts.
- **Explicit instructions**: Liu et al. (2021) show task descriptions help; Wei et al. (2022) show step-by-step reasoning improves few-shot performance.
- **This paper** focuses specifically on the effect of **post-answer** explanations — distinct from chain-of-thought which places reasoning *before* the answer.

(Lampinen et al. 2022)

🧉 Research Questions ---

- Can language models benefit from explanations when learning from examples in-context?
- Do few-shot explanations help the model “understand” the task better?
- What kind of in-context learning abilities do LMs exhibit more generally?

🧉 Contributions & Main Findings ---

- Annotated **40 challenging tasks** from BIG-Bench with human-written explanations.
- Evaluated the effect of **post-answer explanations** (contrast to Wei et al.'s pre-answer chains).

Key findings

- Explanations improve LLM performance compared to matched control conditions.
- Explanations tuned on a small validation set yield even larger gains.

(Lampinen et al. 2022)

🧉 Methods: Data & Model ---

Data: 40 challenging tasks from BIG-Bench, including:

- Inferring goals from actions
- Reasoning about mathematical induction
- Reasoning about causality
- Inferring assumptions behind statements

Model: Decoder-only Transformer LMs (Gopher family: 1B, 7B, 280B parameters), evaluated with the same context window and training data.

🧉 Methods: Explanation Annotation ---

- One author annotated **15 examples per task** with human explanations (restricted to multiple-choice tasks).
- Three **control conditions** to isolate the causal effect of explanations:
  - *True non-explanations*: relevant but non-explanatory text.
  - *Other item explanations*: explanations drawn from a different task example.
  - *Scrambled explanations*: words of the explanation randomly reordered.
- **Explanation fine-tuning**: greedily select best 5-shot prompt on a 10-example validation set; also hand-tune explanation wording.

Methods: Evaluation ---

- Results modeled with **hierarchical logistic regression** to account for:
  - Task difficulty
  - Shared content between prompts
- Each effect estimated at the appropriate level of the hierarchy, quantifying the **unique added contribution** of each prompt component.

How Much Do Explanations Help? ---

🧉 Effect Size (Gopher 280B) ---

Selection vs Untuned Explanations ---

🧉 When Do Explanations Help? ---

🧉 Is Scale Necessary? ---

Not All Tasks Are Created Equal ---

🧉 Discussion ---

Open questions

- Can explanations improve few-shot learning? **Yes, with sufficient scale and quality.**
- Why are post-answer explanations interesting? They decouple reasoning from generation and have no test-time cost.
- What do results imply about in-context learning? LMs can extract semantic content from explanations, not just surface structure.
- How do explanations relate to instructions? They appear complementary.
- How does this relate to human language processing? LMs differ — they lack broader world context.

🧉 Class Discussion ---

- Is the dataset large (~600 samples) / diverse (40 tasks) enough to support the conclusions?
- Is the degree of improvement offered by explanations practically significant?
- How does this compare to other in-context learning methods?
  - Chain-of-thought (Wei, Wang, et al. 2022)
  - Soft prompts / prompt tuning
  - Finetuning + distillation (Alpaca)
- How could explanations benefit smaller models?

Thank You!

🧉 References --- |

Brown, Tom, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. 2020. "Language Models Are Few-Shot Learners." *Advances in Neural Information Processing Systems* 33: 1877–1901.

Chen, Mark, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, et al. 2021. "Evaluating Large Language Models Trained on Code." *arXiv Preprint arXiv:2107.03374*.

Cobbe, Karl, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, et al. 2021. "Training Verifiers to Solve Math Word Problems." *arXiv Preprint arXiv:2110.14168*.

Koncel-Kedziorski, Rik, Subhro Roy, Aida Amini, Nate Kushman, and Hannaneh Hajishirzi. 2016. "MAWPS: A Math Word Problem Repository." In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics*, 1152–7.

Lampinen, Andrew K, Ishita Dasgupta, Stephanie CY Chan, Kory Matthewson, Michael Henry Tessler, Antonia Creswell, James L McClelland, Jane X Wang, and Felix Hill. 2022. "Can Language Models Learn from Explanations in Context?" *arXiv Preprint arXiv:2204.02329*.

🧉 References ---]🧉 References ---

Ling, Wang, Dani Yogatama, Chris Dyer, and Phil Blunsom. 2017. "Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems." *arXiv Preprint arXiv:1705.04146*.

Miao, Shen-yun, Chao-Chun Liang, and Keh-Yih Su. 2020. "A Diverse Corpus for Evaluating and Developing English Math Word Problem Solvers." In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 975–84.

Patel, Arkil, Chitta Baral, and Chitta Baral. 2021. "Are NLP Models Really Able to Solve Simple Math Word Problems?" In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, 2080–94.

Tay, Yi, Mostafa Dehghani, Vinh Q Tran, Xavier Garcia, Dara Bahri, Tal Schuster, Huaixiu Steven Zheng, Neil Houlsby, and Donald Metzler. 2022. "UL2: Unifying Language Learning Paradigms." *arXiv Preprint arXiv:2205.05131*.

Thoppilan, Romal, Daniel De Freitas, Jamie Hall, Noam Shazeer, Apoorv Kulshreshtha, Heng-Tze Cheng, Alicia Jin, et al. 2022. "LaMDA: Language Models for Dialog Applications." *arXiv Preprint arXiv:2201.08239*.

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Wei, Jason, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, et al. 2022. "Emergent Abilities of Large Language Models." *Transactions on Machine Learning Research*.

Wei, Jason, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models." *Advances in Neural Information Processing Systems* 35: 24824–37.